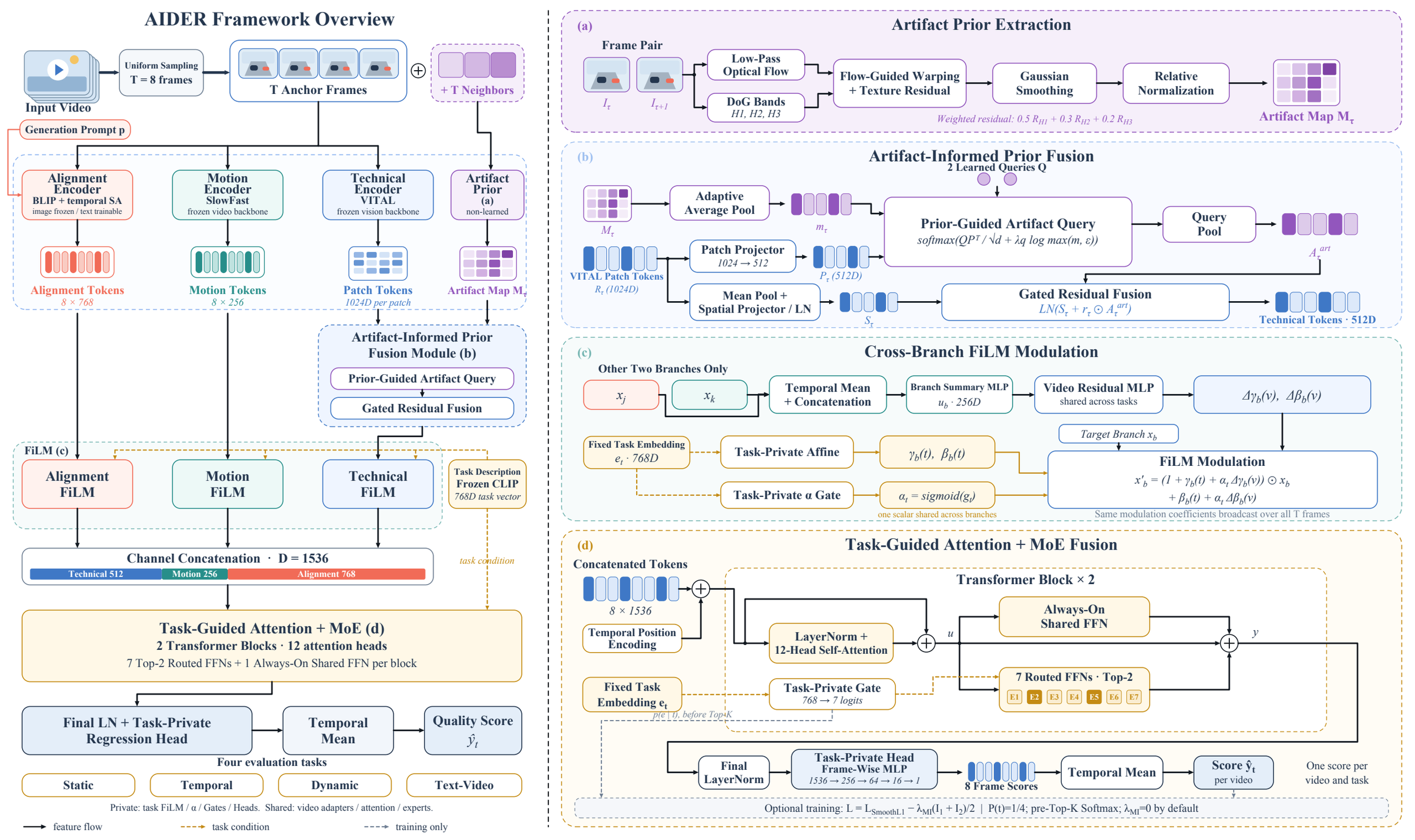

AIDER Framework Overview
Input Video
Uniform Sampling
T = 8 frames
T Anchor Frames
+ T Neighbors
Generation Prompt p
Alignment Encoder
BLIP + temporal SA
image frozen / text trainable
Motion Encoder
SlowFast
frozen video backbone
Technical Encoder
VITAL
frozen vision backbone
Artifact Prior
(a)
non-learned
Alignment Tokens
8 × 768
Motion Tokens
8 × 256
Patch Tokens
1024D per patch
Artifact Map M_τ
Artifact-Informed Prior Fusion Module (b)
Prior-Guided Artifact Query
Gated Residual Fusion
FiLM (c)
Alignment FiLM
Motion FiLM
Technical FiLM
Task Description
Frozen CLIP
768D task vector
task condition
Channel Concatenation · D = 1536
Technical 512
Motion 256
Alignment 768
Task-Guided Attention + MoE (d)
2 Transformer Blocks · 12 attention heads
7 Top-2 Routed FFNs + 1 Always-On Shared FFN per block
Final LN + Task-Private Regression Head
Temporal Mean
Quality Score
ŷ_t
Four evaluation tasks
Static
Temporal
Dynamic
Text-Video
Private: task FiLM / α / Gates / Heads. Shared: video adapters / attention / experts.
feature flow
task condition
training only
(a)
Artifact Prior Extraction
Frame Pair
I_τ
I_τ+1
Low-Pass Optical Flow
DoG Bands
H1, H2, H3
Flow-Guided Warping + Texture Residual
Gaussian Smoothing
Relative Normalization
Weighted residual: 0.5 R_H1 + 0.3 R_H2 + 0.2 R_H3
Artifact Map M_τ
(b)
Artifact-Informed Prior Fusion
2 Learned Queries Q
M_τ
Adaptive Average Pool
m_τ
Prior-Guided Artifact Query
softmax(QP^T / √d + λq log max(m, ε))
Query Pool
A_τ^art
VITAL Patch Tokens
R_t (1024D)
Patch Projector
1024 → 512
P_τ (512D)
Mean Pool + Spatial Projector / LN
S_τ
Gated Residual Fusion
LN(S_τ + r_τ ⊙ A_τ^art)
Technical Tokens · 512D
(c)
Cross-Branch FiLM Modulation
Other Two Branches Only
x_j
x_k
Temporal Mean + Concatenation
Branch Summary MLP
u_b · 256D
Video Residual MLP
shared across tasks
Δγ_b(v), Δβ_b(v)
Fixed Task Embedding
e_t · 768D
Task-Private Affine
γ_b(t), β_b(t)
Task-Private α Gate
α_t = sigmoid(g_t)
one scalar shared across branches
Target Branch x_b
FiLM Modulation
x'_b = (1 + γ_b(t) + α_t Δγ_b(v)) ⊙ x_b
+ β_b(t) + α_t Δβ_b(v)
Same modulation coefficients broadcast over all T frames
(d)
Task-Guided Attention + MoE Fusion
Concatenated Tokens
8 × 1536
Temporal Position Encoding
Transformer Block × 2
LayerNorm + 12-Head Self-Attention
u
Always-On Shared FFN
y
Fixed Task Embedding e_t
Task-Private Gate
768 → 7 logits
p(e | t), before Top-K
7 Routed FFNs · Top-2
E1
E2
E3
E4
E5
E6
E7
Final LayerNorm
Task-Private Head Frame-Wise MLP
1536 → 256 → 64 → 16 → 1
8 Frame Scores
Temporal Mean
Score ŷ_t
per video
One score per video and task
Optional training: L = L_SmoothL1 − λ_MI(I_1 + I_2)/2 | P(t)=1/4; pre-Top-K Softmax; λ_MI=0 by default